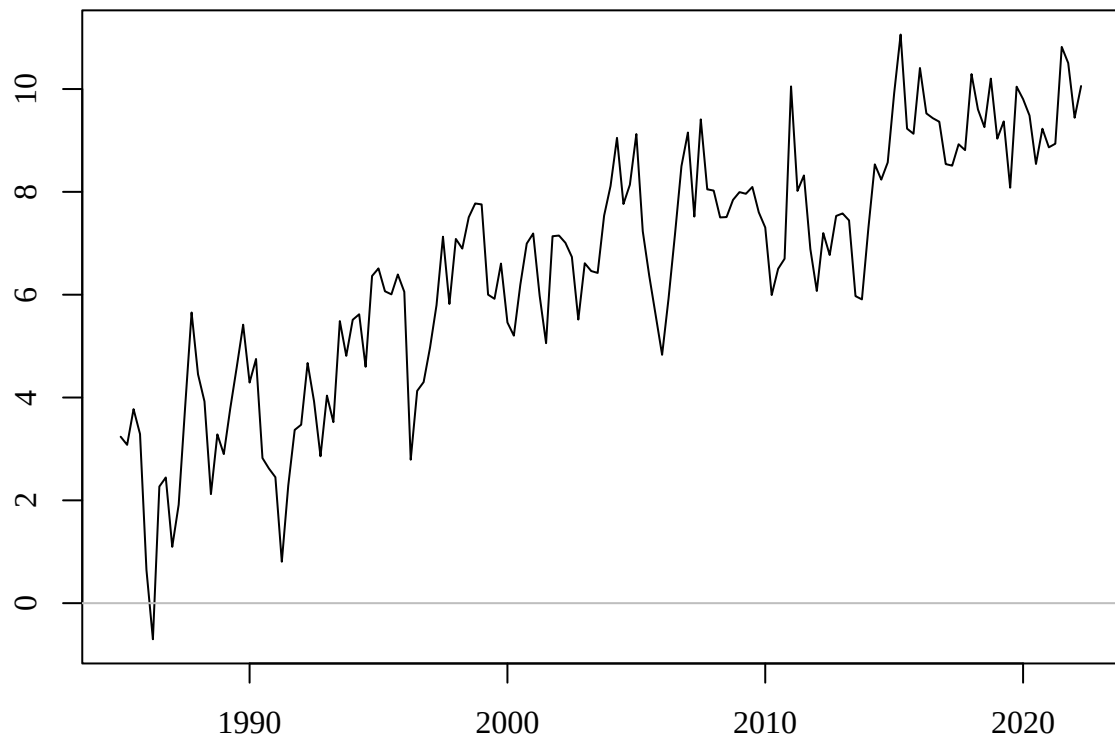


Question 1. (8 marks)

The following is a plot of a quarterly time series from 1985q1 to 2022q2, a total of 150 observations.



- (a) (2 marks) Does this time series plot suggest a **trend** is present? Answer “yes”, “no” or “can’t tell”, with a one sentence explanation.

(b) (2 marks) Does the time series plot suggest a **seasonal pattern** is present? Answer “yes”, “no” or “can’t tell”, with a one sentence explanation.

(c) (2 marks) Does the time series plot suggest **non-stationarity** is present? Answer “yes”, “no” or “can’t tell”, with a one sentence explanation.

(d) (2 marks) Does the time series plot suggest **non-invertibility** is present? Answer “yes”, “no” or “can’t tell”, with a one sentence explanation.

Question 2. (11 marks)

An augmented Dickey-Fuller test regression with trend was calculated for the time series in the plot in Question 1 using the `urca` command `ur.df`:

```
ur.df(Y, type="trend", lags=1)
```

The following test regression was obtained.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.270	0.270	4.702	0.000
z.lag.1	-0.430	0.076	-5.673	0.000
tt	0.021	0.004	5.043	0.000
z.diff.lag	0.040	0.083	0.475	0.636

(a) (2 marks) Write out the estimated test regression in equation form.

(b) (1 mark) What is the value of the augmented Dickey-Fuller t test statistic?

(c) (2 marks) What are the null and alternative hypotheses of the Dickey-Fuller test?

The `urca` command `qunitroot(q, trend="ct")` is used to compute the critical values for the augmented Dickey-Fuller test for $q = 0.05, 0.01, 0.005, 0.001$. The output is shown here:

	5%	1%	0.5%	0.1%
ADF critical value:	-3.41	-3.958	-4.164	-4.596

- (d) (3 marks) Carry out the augmented Dickey-Fuller test at the 5% level of significance using the statistic in part (b). State the test decision and conclusion.

- (e) (3 marks) If we were to compute the p -value for the test statistic in part (b), would this p -value be greater than or less than 0.001? Explain.

If you run out of space for Question 2, use the extra pages at the end of the exam paper. Please clearly indicate there the question number/part you are answering.

☐ Please tick this box if you used extra pages for any part of Question 2.

Question 3. (23 marks)

For a stationary time series with $n = 200$ observations a range of $\text{ARMA}(p, q)$ were fitted to attempt to find a suitable forecasting model. The following table shows the values of the AICc and p -values of the Ljung-Box residual autocorrelation test for every combination of $p = 0, 1, 2, 3$ and $q = 0, 1, 2, 3$.

p	q	AICc	LBp
0	0	240.81	0.000
1	0	99.76	0.008
2	0	99.95	0.725
3	0	101.32	0.608
0	1	129.60	0.000
1	1	99.84	0.937
2	1	101.36	0.910
3	1	103.44	0.609
0	2	103.75	0.313
1	2	101.20	0.997
2	2	103.30	0.948
3	2	105.33	0.667
0	3	101.86	0.915
1	3	103.29	1.000
2	3	105.44	0.950
3	3	99.98	0.723

- (a) (3 marks) According to the AICc alone, what are the best three models? (List in order of first, second, third best.) Explain.

- (b) (5 marks) Taking into account the Ljung-Box test results (at the 5% level) in addition to the AICc from part(a), which model would be selected as overall “best”? Explain how and why the Ljung-Box test results are relevant to this selection.

The coefficient estimates of the ARMA(1,1) and ARMA(2,0) models are as follows:

```
          ar1    ma1 intercept
ARMA(1,1): 0.523 0.355      2.751

          ar1    ar2 intercept
ARMA(2,0): 0.874 -0.263      2.747
```

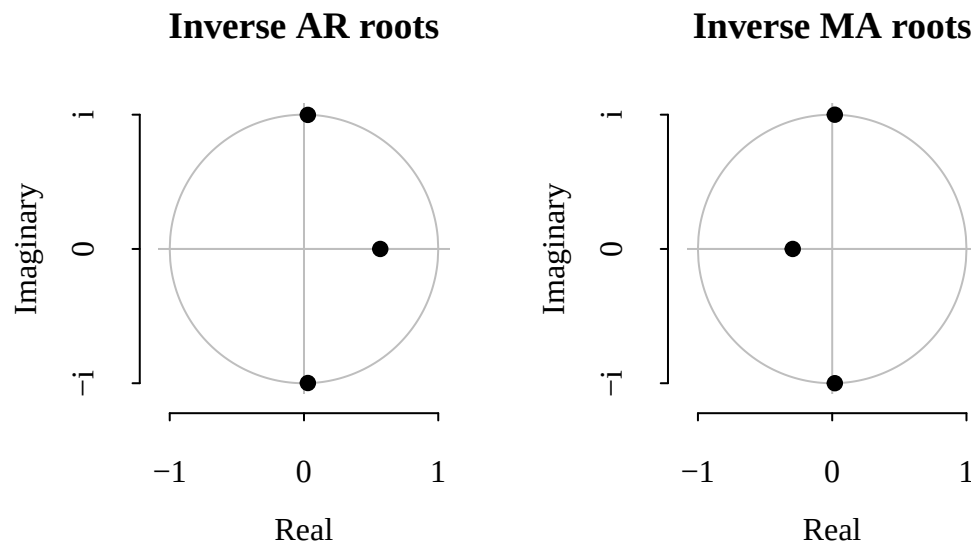
- (c) (4 marks) Write out both models in equation form.

Forecasts from the ARMA(1,1) and ARMA(2,0) models for $h = 1, 2, \dots, 8$ steps ahead are tabulated here:

h	ARMA(1,1)	ARMA(2,0)
1	3.165	3.089
2	2.967	2.851
3	2.864	2.748
4	2.810	2.720
5	2.781	2.723
6	2.767	2.733
7	2.759	2.741
8	2.755	2.745

- (d) (6 marks) The two models have quite different functional forms and coefficient estimates, but quite similar forecasts. Give an explanation for this.

Referring to the table of AICc values, the ARMA(3,3) model may be another candidate model to try. The estimation of this model produces the following plot of its inverse roots.



- (e) (5 marks) Give an explanation of what information these two plots suggest about the specification of the ARMA(3,3) model.

If you run out of space for Question 3, use the extra pages at the end of the exam paper. Please clearly indicate there the question number/part you are answering.

☐ Please tick this box if you used extra pages for any part of Question 3.

Question 4. (12 marks)

We have $n = 300$ observations on two time series, $Y_{1,t}$ and $Y_{2,t}$, to use as estimation sample, with the following $h = 5$ observations retained for forecast evaluation purposes. Following specification searches, the following models are fitted:

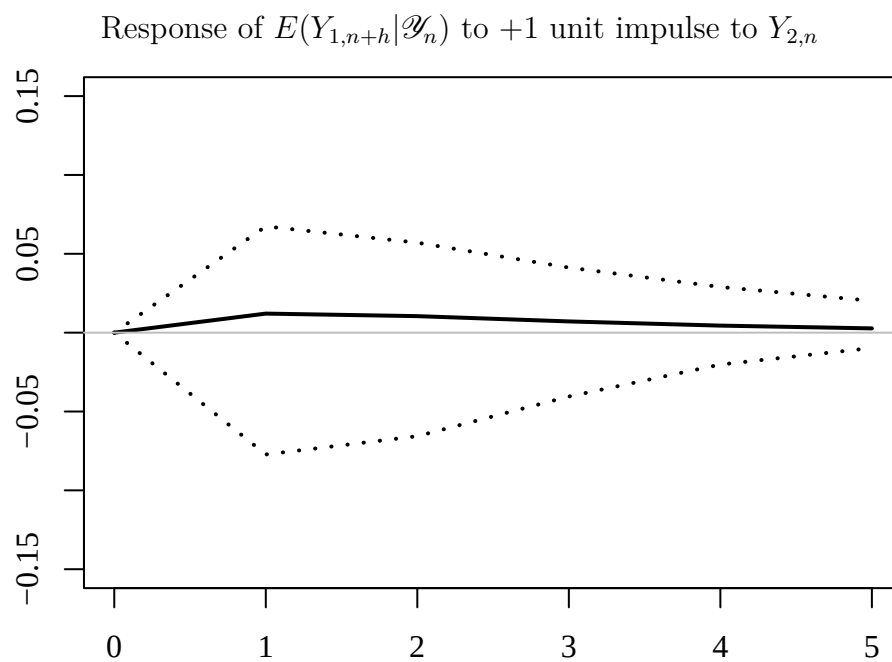
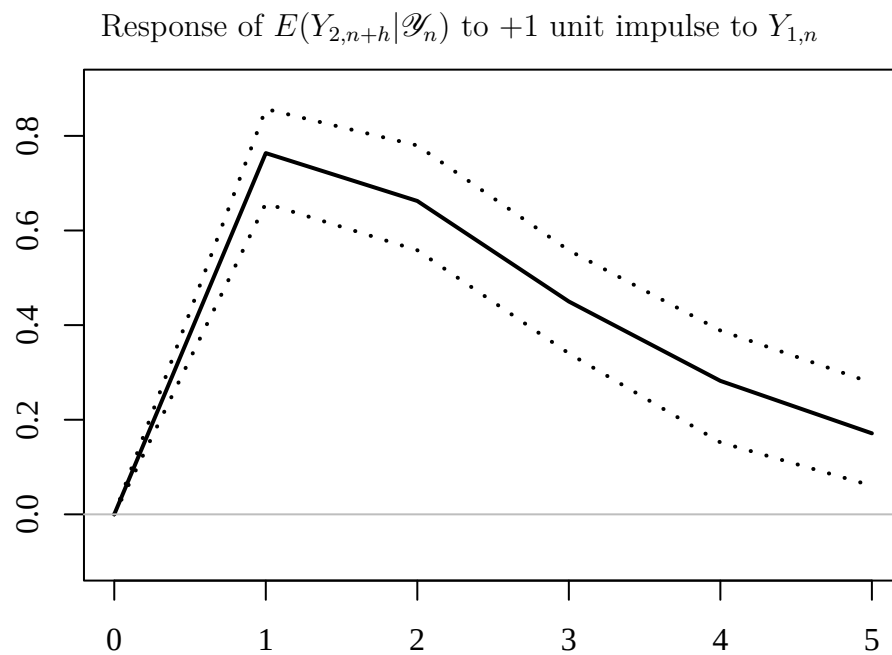
- univariate AR(1) model for $Y_{1,t}$,
- univariate AR(2) model for $Y_{2,t}$,
- bivariate VAR(1) model for $Y_{1,t}$ and $Y_{2,t}$.

Forecasts of $Y_{1,t}$ and $Y_{2,t}$ were generated from these three models for $t = n + 1, \dots, n + 5$, and Root Mean Squared Errors calculated. The following table shows the RMSEs for the forecasts of $Y_{1,t}$ (Y1) and $Y_{2,t}$ (Y2) from the corresponding univariate AR model (AR) and the bivariate VAR(1) model (VAR).

	Y1	Y2
VAR	0.734	0.908
AR	0.703	0.993

- (a) (4 marks) Explain what these results imply about the presence or absence of Granger causality in either direction between $Y_{1,t}$ and $Y_{2,t}$.

Using the VAR(1) model for $Y_{1,t}$ and $Y_{2,t}$, *non-orthogonalised* impulse response functions were calculated and plotted below, along with 95% confidence intervals (in dashed lines).



If you run out of space for Question 4, use the extra pages at the end of the exam paper. Please clearly indicate there the question number/part you are answering.

☐ Please tick this box if you used extra pages for any part of Question 4.

(b) (4 marks) Interpret these impulse response functions.

(c) (4 marks) In addition discuss whether the implications of the impulse responses are consistent or inconsistent with the Granger causality findings in part (a).

Question 5. (14 marks)

Consider a model for a quarterly time series Y_t :

$$Y_t = \beta_0 + \beta_1 Q_{1,t} + \beta_2 Q_{2,t} + \beta_3 Q_{3,t} + Z_t$$

$$Z_t = 1.3Z_{t-1} - 0.4Z_{t-2} + U_t + 0.5U_{t-1},$$

where U_t is the 1-step-ahead prediction error and $Q_{1,t}$, $Q_{2,t}$ and $Q_{3,t}$ are seasonal dummy variables taking the value 1 in quarters 1, 2 and 3 respectively, and the value 0 otherwise.

(a) (2 marks) Express the equation for Z_t using lag operators and polynomials.

(b) (4 marks) Is it possible that the model for Z_t is stationary? Explain why or why not, and why the question needs to be qualified by "is it possible".

(c) (2 marks) Is the model for Z_t invertible? Explain why or why not.

(d) (2 marks) Does the model for Z_t contain a common factor? Explain why or why not.

(e) (4 marks) Is it possible that the entire model for Y_t (i.e. both equations considered together) is stationary? If necessary, explain any special case(s) that affect your general answer.

If you run out of space for Question 5, use the extra pages at the end of the exam paper. Please clearly indicate there the question number/part you are answering.

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Please tick this box if you used extra pages for any part of Question 5.

Question 6. (22 marks)

Let Y_t be a time series that satisfies the following AR(1)-ARCH(1) model

$$Y_t = \beta_0 + Z_t$$

$$Z_t = \phi_1 Z_{t-1} + U_t$$

$$E(U_t | \mathcal{Y}_{t-1}) = 0$$

$$E(U_t^2 | \mathcal{Y}_{t-1}) = \omega + \alpha_1 U_{t-1}^2.$$

- (a) (2 marks) Is it valid to consider U_t as the *one-step-ahead prediction error* in this model when it has the ARCH(1) structure?

- (b) (2 marks) Why is it required to impose $\omega > 0$ and $\alpha_1 >$ in this model?

(c) (6 marks) What parameter restriction is required for U_t to be stationary? Under this restriction, derive the *unconditional* mean and variance of U_t .

- (d) (4 marks) Give expressions for the conditional mean and conditional variance of Y_t , i.e. $E(Y_t | \mathcal{Y}_{t-1})$ and $\text{var}(Y_t | \mathcal{Y}_{t-1})$, as functions of lags of Y_t .

- (e) (2 marks) How would such expressions for $E(Y_t | \mathcal{Y}_{t-1})$ and $\text{var}(Y_t | \mathcal{Y}_{t-1})$ be useful for constructing a normal 95% prediction interval for the one-step-ahead forecast $E(Y_{n+1} | \mathcal{Y}_n)$?

(f) (2 marks) Referring to your answer to (d), which lags of Y_n are used in computing the prediction interval in the AR(1)-ARCH(1) model in this question?

(g) (2 marks) Referring to your answer to (d), how does the value of the one-step-ahead forecast for Y_{n+1} relate to the value of the ARCH(1) coefficient (α_1)?

(h) (2 marks) Referring to your answer to (d), how does the width of the one-step-ahead prediction interval for Y_{n+1} relate to the value of the ARCH(1) coefficient (α_1)?

End of questions.

Extra page. Clearly indicate each question number/part you are answering.

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